Case H — Mr. David Thompson | 55M | Chronic Bronchitis (Type B COPD — "Blue Bloater")

Bus driver. Productive cough ≥3 months/year for 2+ consecutive years. Bluish lips, bilateral peripheral oedema lower legs. High BMI. Smokes 15 cigarettes/day. Currently congested. Seeking airway clearance advice and smoking cessation support to reduce sick leave.

Cardiorespiratory Physiotherapy

■ SUBJECTIVE PROBLEM LIST

- Physical Impairments
- Airflow obstruction (reduced FEV1/FVC on spirometry)
- Reduced exercise tolerance and exertional dyspnoea

■■ RED FLAGS

- ■ SpO2 <88% at rest — significant hypoxaemia / possible type 2 respiratory failure
- In a "blue bloater," chronic CO2
- retention means the hypoxic drive may be sustaining respiratory effort.

■ PATHOPHYSIOLOGY (KEY POINTS)

- Chronic bronchitis results from sustained inhalation of irritants — primarily cigarette smoke — triggering a chronic inflammatory response in the large central airways.
- This causes hypertrophy and hyperplasia of the submucosal mucous glands, quantified by the Reid Index (ratio of gland layer thickness to total bronchial wall thickness — normal <0.4; chronic bronchitis >0.4).
- Goblet cell metaplasia also extends distally into smaller airways that do not normally secrete mucus.
- Together, these changes produce dramatically increased volumes of thick, viscid mucus that overwhelm the mucociliary escalator, leading to mucus pooling, bacterial colonisation, and recurrent lower respiratory tract infections.
- Chronic inflammation, airway wall oedema, and mucus plugging narrow the lumen, producing the obstructive spirometric pattern (reduced FEV1/FVC) characteristic of COPD.
- Why this patient fits: Mr.

■ DIFFERENTIAL DIAGNOSES

See full case notes

■ OBJECTIVE ASSESSMENTS & HOW TO PERFORM

Assessment & Supervised Trial of ACBT (Active Cycle of Breathing Technique)

What it measures: ACBT is the first-line, evidence-based airway clearance technique for chronic bronchitis. The supervised trial assesses the patient's ability to perform each phase, evaluates immediate sputum production and clearance,

baseline; yellow/green = infection) and volume. Failure to clear with

ACBT alone may indicate need for oscillating PEP (Acapella, Flutter valve) to break up tenacious secretions. I select this test because: "ACBT directly addresses Mr. Thompson's primary complaint — congestion — and

"Auscultation is quick, non-invasive, and guides everything else. It

confirms whether Mr. Thompson's presentation is consistent with mucus in large airways (supporting ACBT) or whether there is consolidation or effusion that would change my approach and require urgent referral. Repeating it immediately post-ACBT lets me demonstrate to Mr. Thompson — and to the examiner — that the technique has worked, which reinforces

■ TREATMENT PLAN

- that reducing infections through effective airway clearance and smoking
- cessation is a clinically meaningful and achievable target for Mr.
- 6. "On a scale of 0 to 10, how ready are
- you to make a change to your smoking right now — and what feels like the
- biggest barrier for you?"
- I ask this because it assesses readiness to
- change using the Motivational Interviewing framework and Prochaska's
- Stages of Change model. The answer tells me whether to plant information
- seeds (precontemplation), explore ambivalence (contemplation), or move
- into action planning (preparation stage). I do not want to push a quit
- plan on someone not ready — it reduces engagement and damages
- therapeutic alliance.